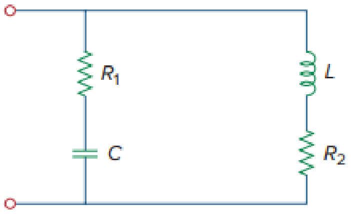
Problem

Using Fig. 9.65, design a problem to help other students better understand impedance combinations.

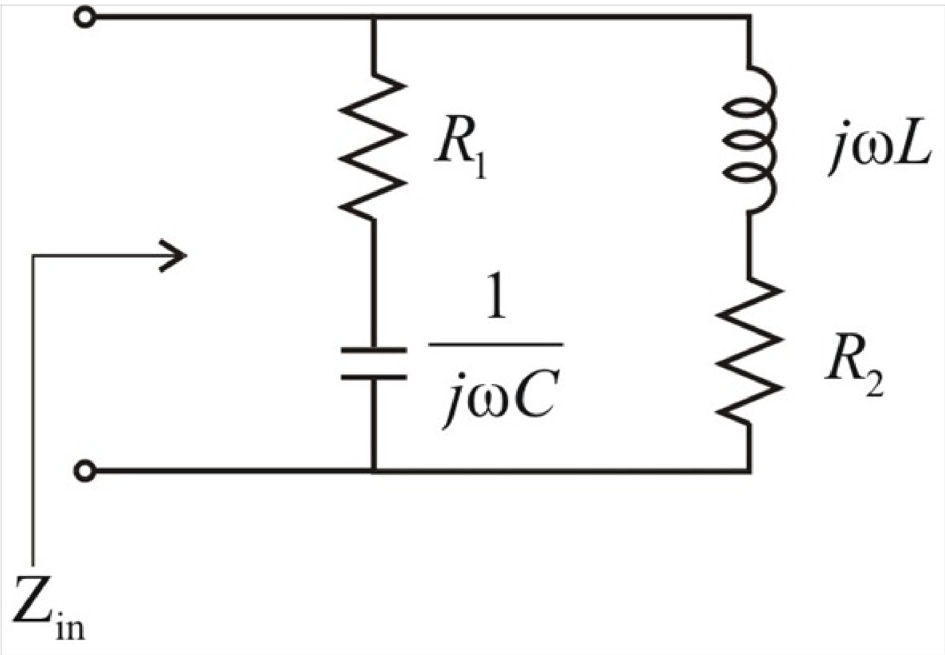
Figure 9.65:



Step-by-step solution

Step 1 of 2

Given circuit can be represented as



Step 2 of 2

From the above circuit,

$$Z_{in} = \left(R_1 + \frac{1}{j\omega C}\right) \Big\| \left(j\omega L + R_2\right)$$

$$Z_{in} = \left(\frac{j\omega R_1 C + 1}{j\omega C}\right) \Big\| \left(j\omega L + R_2\right)$$

$$Z_{in} = \frac{\frac{\left(j\omega R_1 C + 1\right)}{j\omega C} \times \left(j\omega L + R_2\right)}{\left(\frac{j\omega R_1 C + 1}{j\omega C}\right) + \left(j\omega L + R_2\right)}$$

$$Z_{in} = \frac{\left(j\omega R_1 C + 1\right)\left(j\omega L + R_2\right)}{\left(j\omega R_1 C + 1\right) + \left(j\omega L + R_2\right)\left(j\omega C\right)}$$

$$Z_{in} = \frac{\left(-\omega^2 R_1 LC + j\omega L + j\omega R_1 R_2 C + R_2\right)}{j\omega R_1 C + 1 - \omega^2 LC + j\omega R_2 C}$$

$$\therefore \boxed{Z_{in} = \frac{j\omega\left(L + R_1 R_2 C\right) - \omega^2 R_1 LC + R_2}{1 - \omega^2 LC + j\omega C\left(R_1 + R_2\right)}}$$